



PHAT-Sat Mission Definition Review

November 12, 2025

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 - Former Industry Advisor, Mike Wieggers Department of Electrical and Computer Engineering, Kansas State University
- **Chris Hersman**
 - Mission Systems Engineer, John Hopkins University Applied Physics Laboratory
- **Dr. Tim Steffensmeier**
 - Assistant Vice President and Director of Engagements and Outreach, Kansas State University
- **Dr. Amir Bahadori**
 - Professor and Program Director, Kansas State University Nuclear Engineering
 - Former NASA civil servant in the Space Radiation Analysis Group (SRAG)



Mission Advisory Board

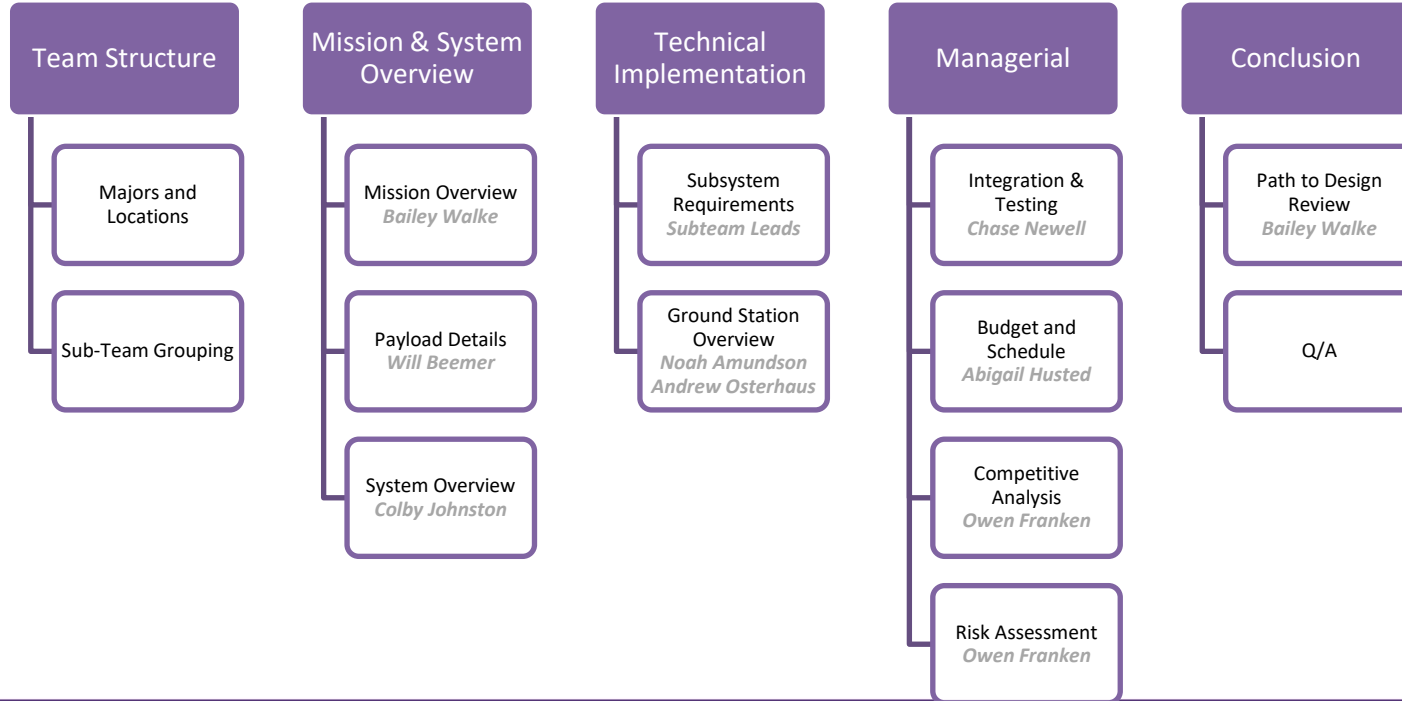
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 - Dean and Professor, Carl R. Ice College of Engineering, Kansas State University
 - Institute for Digital Agriculture and Advanced Analytics (ID3A), Kansas State University
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 - Space Communications System Engineer, John Hopkins University Applied Physics Laboratory
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 - Professor of Geography and Geospatial Sciences, Kansas State University
 - Director, Geographic Information Systems Spatial Analysis Laboratory (GISSAL) and Institute of Digital Agriculture and Advanced Analytics (ID3A)
- **Bill Swartz**
 - Atmospheric and Remote Sensing Scientist, John Hopkins University Applied Physics Laboratory



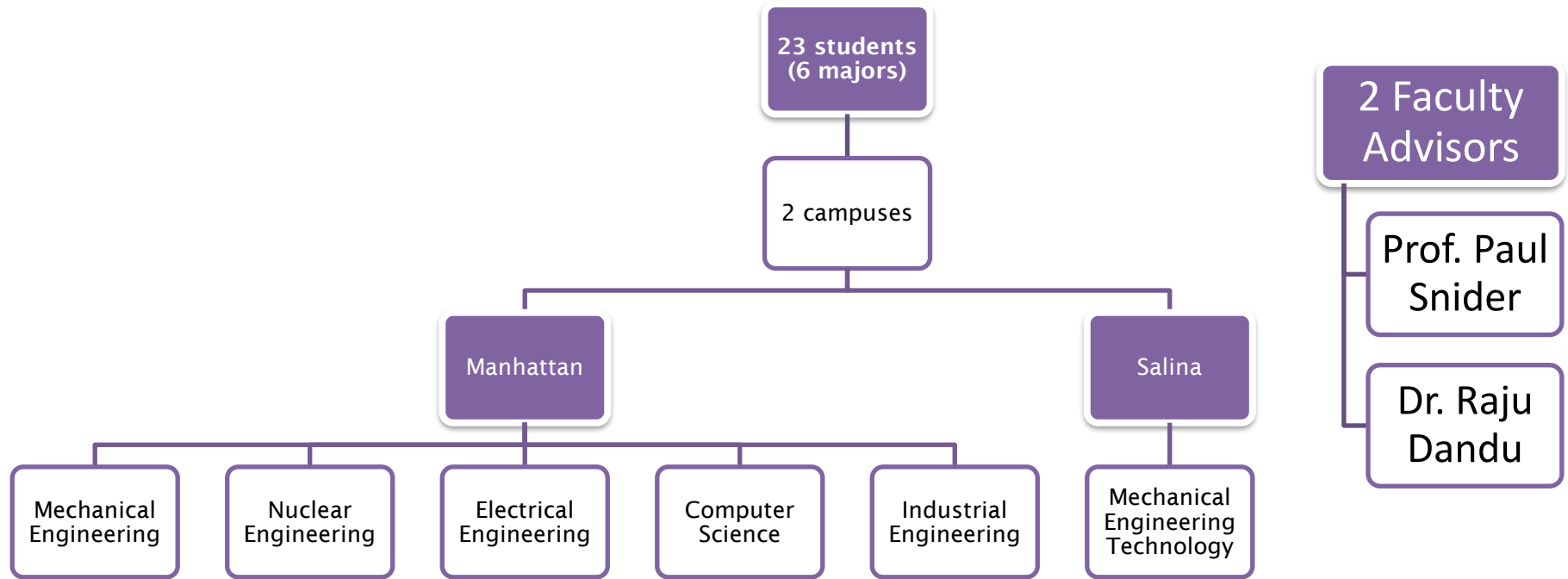
CubeSat Project Donor

- Kevin Needham
 - President, Kiewit Power Engineering
 - Senior Vice President, Power Markets & Strategy

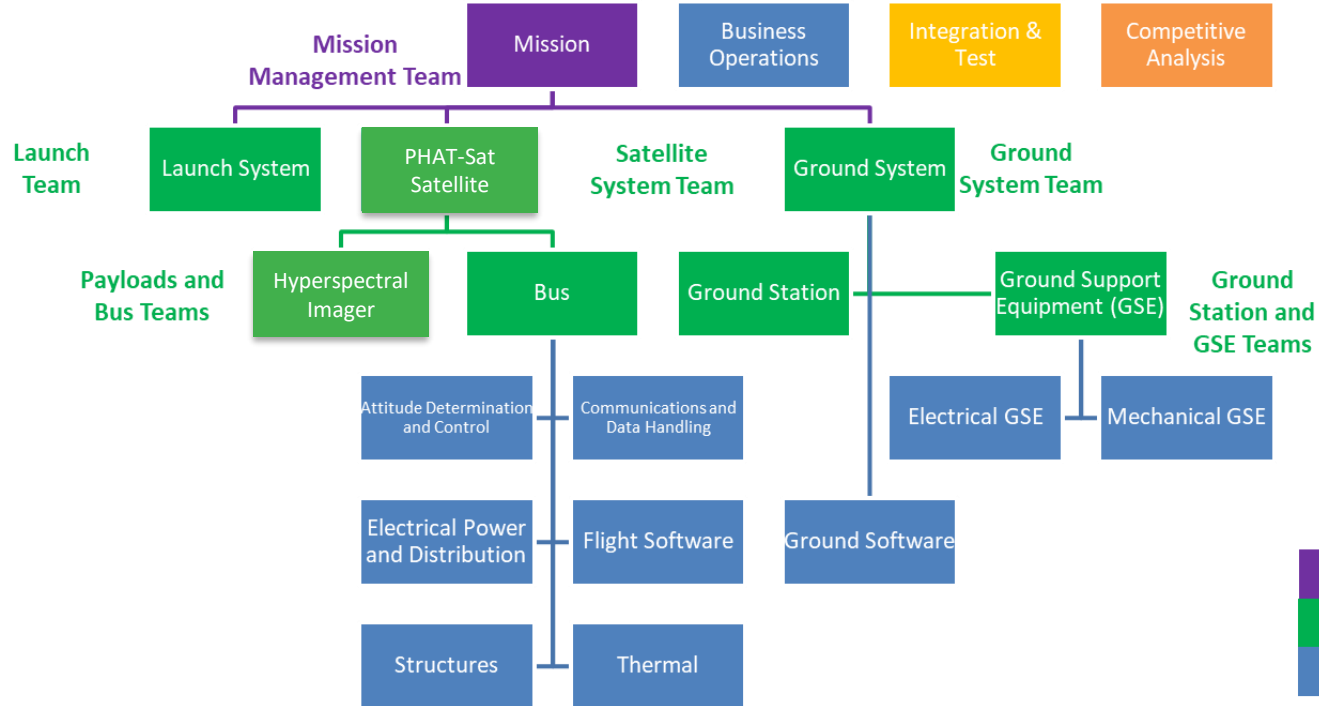
Agenda



Team Acknowledgement



Team Structure





Purpose of MDR

Objective: Define mission concepts, objectives, and system-level requirements; Verify subsystems and operation concepts align with mission goals

Outcomes

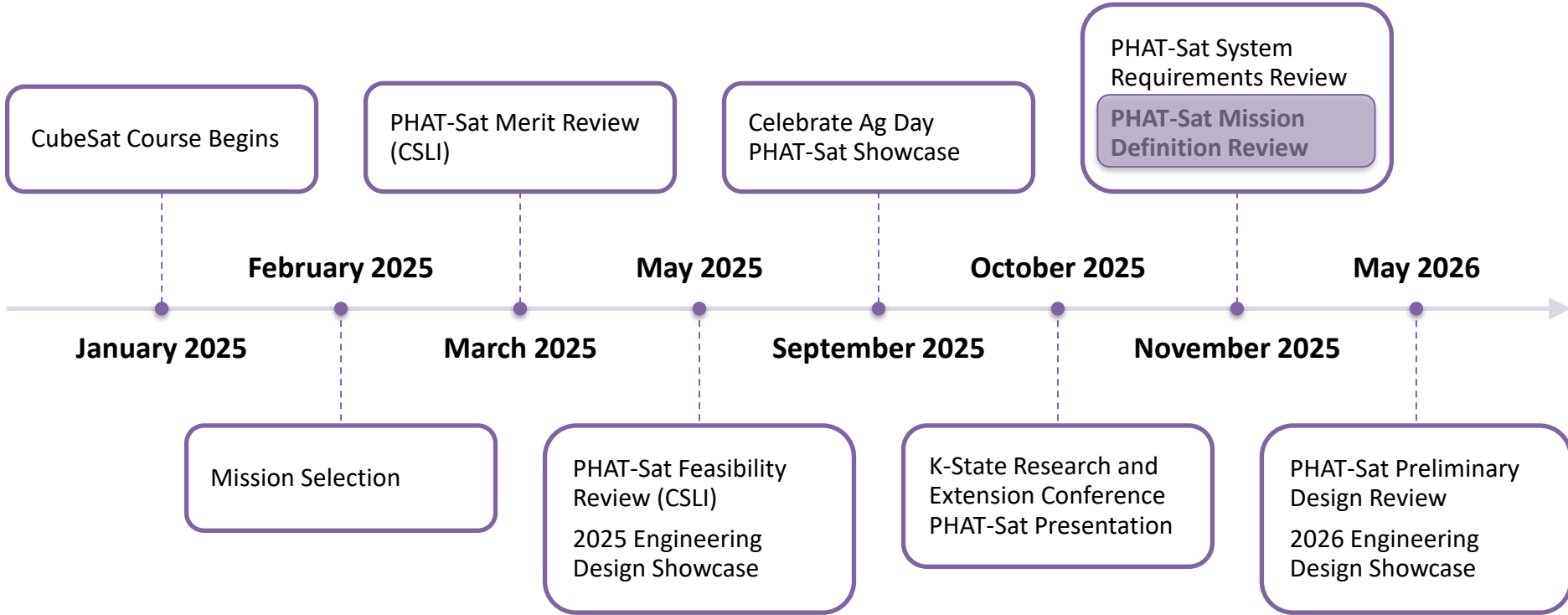
- Confirmation of mission feasibility
- Definition of mission requirements and architecture
- Identification of major risks and mitigation strategies

Review Focus Areas

- Mission objectives and rationale
- System concept and interfaces
- Orbit and environment assumptions
- Schedule, budget and risk strategies
- Readiness for path forward to design review

Not a Design Review; No satellite design details provided (i.e. No quantitative requirements, specifications, budgets, drawings)

PHAT-Sat Timeline



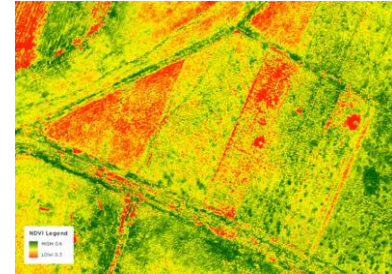


MISSION OVERVIEW

Bailey Walke, Will Beemer

Mission Background

- Kansas Agriculture: \$76 billion total economic contribution ¹
 - 60-70% of gross profits were spent on farm-related expenses
- Feeding a growing population and reducing environmental impact and input waste
 - Fertilizer, irrigation and pesticide use
 - Drought, soil degradation and disease
- Precision agriculture allows for data-driven decisions on resource management





Current Solutions

- Imaging Satellite Constellations:
 - PlanetScope: 1 day revisit, 4m GSD, MS
 - EOS Sat: 1 day revisit (planned), 0.75m GSD, MS
 - Wyvern Dragonette: 1 day revisit (planned), 5.3m GSD, HS

Subscription-based model, pay for data per acre, 1 day revisit is unnecessary for practical applications

- Drones/UAVs
 - Scheduled flights, 2-20 cm GSD, MS/HS

Inconsistent imaging, costly operations, small imaging area, useful for point checks identified from satellite data

Mission Overview

PHAT-Sat: Precision Hyperspectral Agricultural Tracker

- Miniaturized hyperspectral imaging
 - Increased spectral band information: **20-40 bands**
- Cost-effective revisit rate: **7-8 days**
- Comparable spatial resolution: **4.75m-30m**
- Supporting existing open-source datasets (Landsat-8/-9, Sentinel-2, MODIS)



Primary Goal: Enhance agricultural productivity and sustainability in Kansas by addressing challenges related to crop stress, soil degradation, resource management, and climate variability.



Mission Objectives

ID	Title	Description
PHAT-MO-01	Design, Build, and Launch a Cube-Sat	PHAT-Sat will provide an opportunity for K-State students to successfully design, build, test and launch a CubeSat , from the mission development stage through the launch and mission operations.
PHAT-MO-02	Agricultural Vegetation Health Sensing	PHAT-Sat will use remote sensing to study vegetation health for agricultural purposes, research, related industries.
PHAT-MO-03	Develop SmallSat Technology Capabilities	PHAT-Sat will provide an opportunity to develop small scale technologies for spacecraft .
PHAT-MO-04	STEM Engagement and Outreach	PHAT-Sat will provide educational value to the general public on Cube-Sats and the satellite design and mission cycle, as well as agricultural efficiency and farm resource management .

Solution Product

Satellite: Level 1B Data (NASA EOSDIS SDPs)

- Atmospheric, radiometric and geometric corrections
- Time-referenced, geo-referenced

Ground: Level 3A Data

- Vegetation data calculated and plotted on uniform space-time grid scales
- Periodic summary, per revisit

Level 1B: Research

- Customized VIs using hyperspectral data
- Historical trends
- Space-to-ground comparison

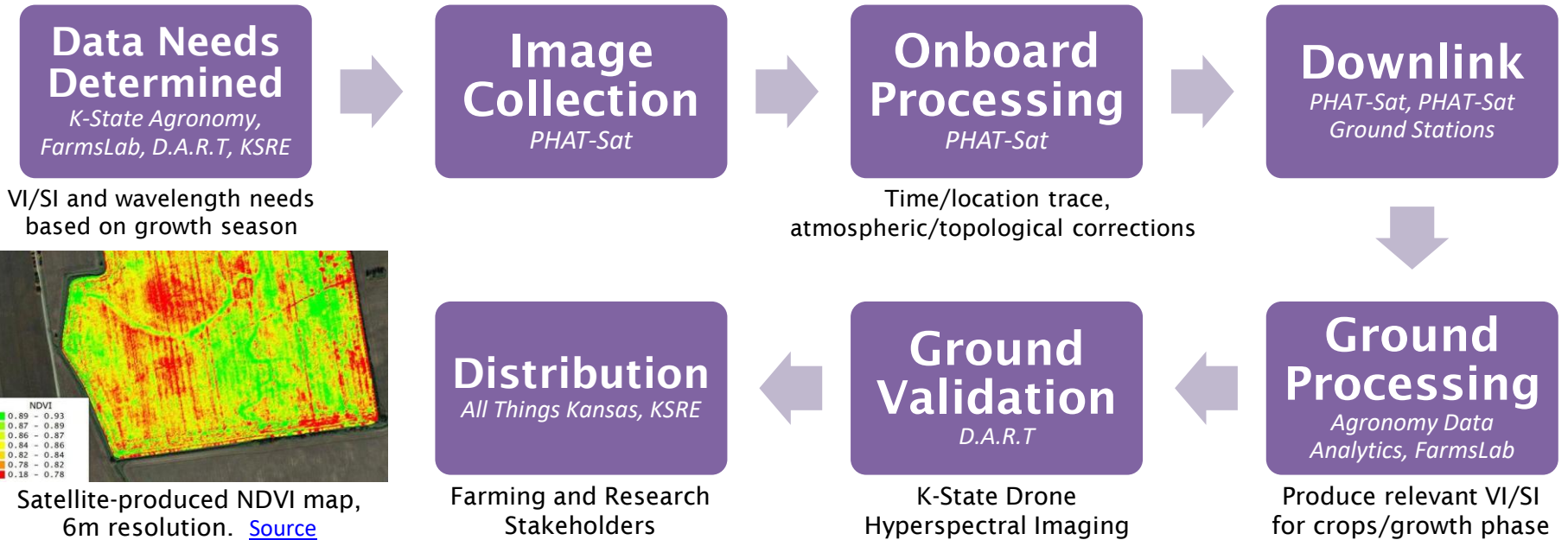
Level 3A: Precision Agriculture

- Easily-interpreted VI heat maps for resource management
- Highlighted anomalies between passes
- Triggers drone or field agent deployment

Index	Purpose
Normalized Difference Vegetation Index	Used to assess overall plant health and biomass
Enhanced Vegetation Index	More accurate measurement of dense vegetation or high atmospheric disturbances
Normalized Difference Water Index	Measures moisture content in vegetation and soil to assess drought conditions
Plant Senescence Reflectance Index	Used to detect early signs of plant senescence and stress
Soil-Adjusted Vegetation Index	Reduces the effect of soil background in areas with low vegetation cover
Modified Soil-Adjusted Vegetation Index	Minimizes effect of soil in areas with low vegetation cover and provides better sensitivity

Relevant vegetation indices (VIs) and their applications

Solution Delivery



Entire product life cycle handled by Kansas State University entities



Progress Since Last Review

Identifying Knowledge Gaps and Who Can Support

- Optics/Hyperspectral Imaging

K-State Geographic Information Science and Technology, D.A.R.T

- Data needs, processing and analysis

Department of Agronomy: D.A.R.T, FarmsLab, Precision Agriculture, Remote Sensing, Data Analytics, KSRE

- Data distribution

All Things Kansas, Department of Agronomy, KSRE

Analyzing Industry Competitors

- Researching private imaging satellites and other remote sensing options

- Understanding stakeholder needs

- Competitive analysis

Understanding of Mission

- Science traceability and imager selection

- System and subsystem requirements definition

- Updated concept of operations

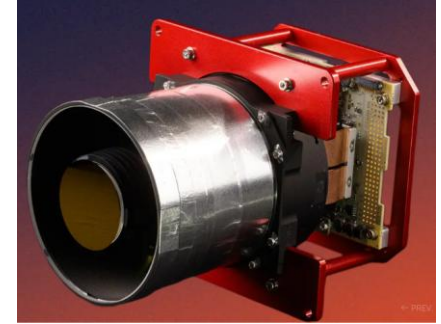
- Risk identification

Science Traceability Matrix

PHAT-Sat Science Traceability Matrix							
Science Goal	Measurement Objective	Measurement Requirement	Instrument	Instrument Requirement		Data Product	Mission Requirement
To provide a continuously updating NDVI map of Kansas that supports precision agriculture for farmers and research for the university's Agriculture Department.	Acquire frequent hyperspectral imagery of Kansas cropland throughout the growing season.	Acquire surface reflectance across $\geq 100\%$ of Kansas cropland per revisit time with a resolution of ≥ 5 pixels per acre	Hyperspectral Imager	Spectral Range	450-850nm	VI moasic map of Kansas	SSO Orbit ≈ 550 km
	Calculate NDVI and other vegetation indices to monitor crop health and detect changes over time.	Insure temporal consintency to compare vegetation indices across a revisit time of ≤ 9 days		Spectral Resolution	≤ 25 nm (optimally 5nm)	Time Series layer stack	downlinking of ≥ 75 GB per revisit
	Deliver processed NDVI maps to the public and research partners for analysis and field validation.	Produce atmospherically corrected surface reflectance and vegetation indices with total uncertainty $\leq XX$		Spatial Resolution	≤ 30 m GSD		Attitude Control $\leq XXX$ deg/s
				Spatial coverage	≤ 122 km		Geolocating tolerancce $\leq XXX$
				Temporal Resolution	$\leq X$ days revisit		Power $\geq XXX$ W sustained

Payload Details

Option	1	2	3
Imager	Simera HyperScape 100	Cosine HyperScout S	Scanway SOP 120
GSD (m)	4.75	28	Customizable
Swath (km)	19.4	115	Customizable
Spectral Range (nm)	460-860	450-950	400-1000
Spectral Resolution (nm)	3% FWAHM (~23nm)	16	~
Size (U)	1.5	1.6	1
Weight (kg)	1.26	1.6	2.5
Peak Power (W)	7	9	6
Total Data (GB)	1882	53.6	50-2000



Simera HyperScape100, Simera Sense



RGB

NDVI

30m GSD
16 sq miles



Sources: [Simera Sense](#) , [Cosine](#) , [Scanway Space](#)



SYSTEM OVERVIEW

Colby Johnston

System Requirements/Mission Block Diagram

Requirements

Image the Midwest
interest area

Geographical and space
location data

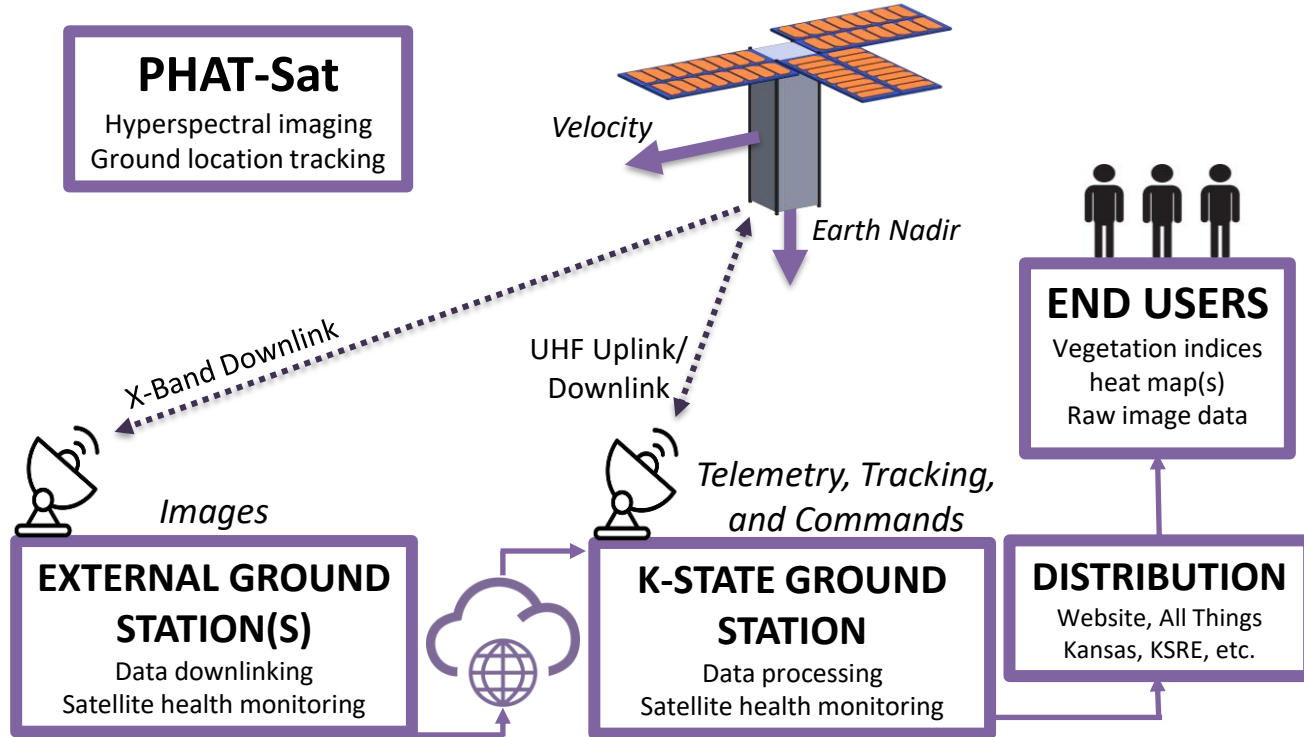
Survives launch and on-orbit
environments

Uplink/Downlink capability
with ground stations

Operational optimization and
data storage safety margin

PHAT-Sat

Hyperspectral imaging
Ground location tracking



Concept of Operations

LAUNCH & INITIALIZATION

1 Month

Deployment from LV



Sleep



Initial Startup



Detumble



Appendages Deploy



Calibration



Stand By

MISSION OPERATIONS

See Following Slide

DECOMISSIONING

6-12 Months

Increased Drag Positioning



Final Transmission



Loss of Communication



Burn Up

MISSION OPERATIONS

24 Months

Satellite

Recurring Science

Imaging

Pointing

Transmitting

Onboard
processing

Telemetry

Lossless data
compression

Periodic/As-Requested

Imager
CalibrationFirmware
Updates

Full Raw Data Transmitting

Safe/Upset

Detumble

EMI Survival

Over/Under
VoltageOver/Under
Temperature

UHF

UHF

X-Band

Ground Stations

Recurring Duties

VIs Creation

Data
DistributionSoftware/Firm
ware UpdatesGround Data
Validation

PHAT-Sat “Day in the Life”

Idle

Charging, ADCS/Imager Calibration
[Prioritize solar charging]

Imaging

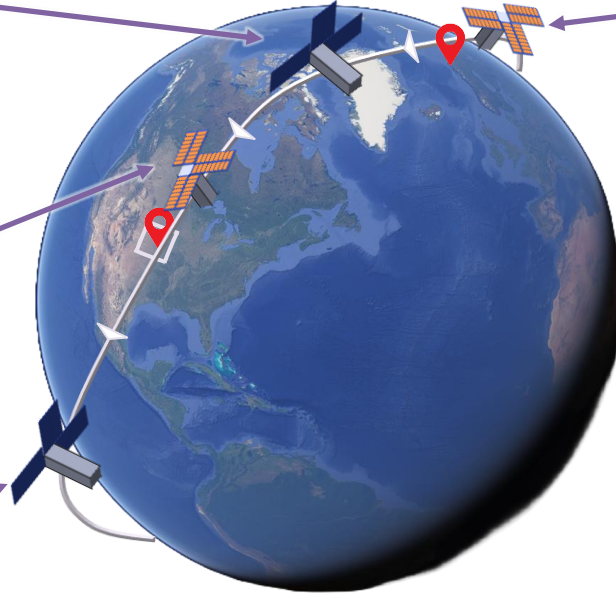
Image all of KS* and parts of
neighboring states**
[Prioritize nadir pointing]

Processing

Data processing and compression
[Prioritize solar charging]

Transmitting

X-band image, UHF health
[Prioritize ground station pointing]



Ground stations:

- 📍 K-State Radio Station
Manhattan, KS (UHF)
- 📍 KSAT (X-Band and UHF)
Svalbard, Norway

Others may be added as needed



ORBIT & ENVIRONMENT

Igor Sheshukov, Phillip Shirkey

Orbit

K-State Manhattan GS Access

Downlink Time 2min-8.3min-11min

Revisit Time 1hr-4.5hr-11.2hr

KSAT Norway GS Access

Downlink Time 1.3min-9.4min-11.25min

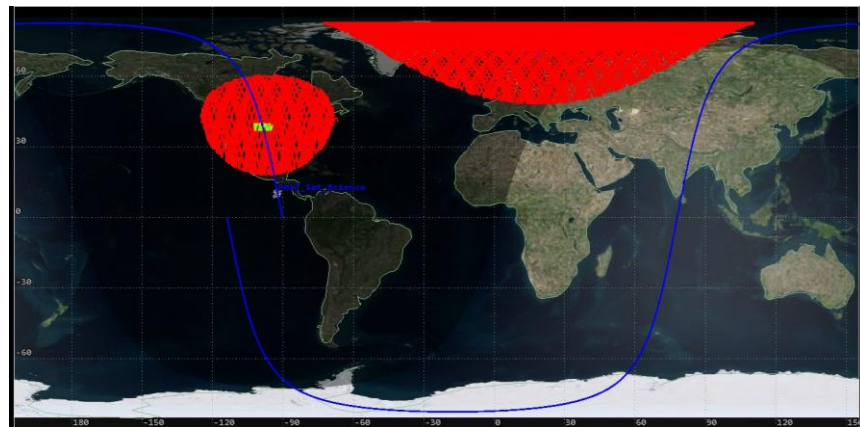
Revisit Time 0.8hr-1.9hr-7.9hr

Kansas State Area Access

Downlink Time 11sec-43sec-50sec

Revisit Time 0.5day-1.3day-2.5day

- Sun Synchronous (SSO)
- ~500km Altitude
- ~97° Inclination
- Lifetime: 3.5 years





Space Environment

- Thermal Cycling (Similar to NuKat)
- Radiation (Higher)
- Debris (Similar)
- Solar and Electromagnetic Events (Higher)
- Atmosphere (Similar*)



Launch Environment

- Shock & Vibrations (Similar)
- Acceleration (Similar)
- Electromagnetic Compatibility (Similar)
- Ascent Venting (Lower)
- Other Deployment Constraints (Higher)
 - Nested deployment risks, imager vibration



SATELLITE SUBSYSTEMS

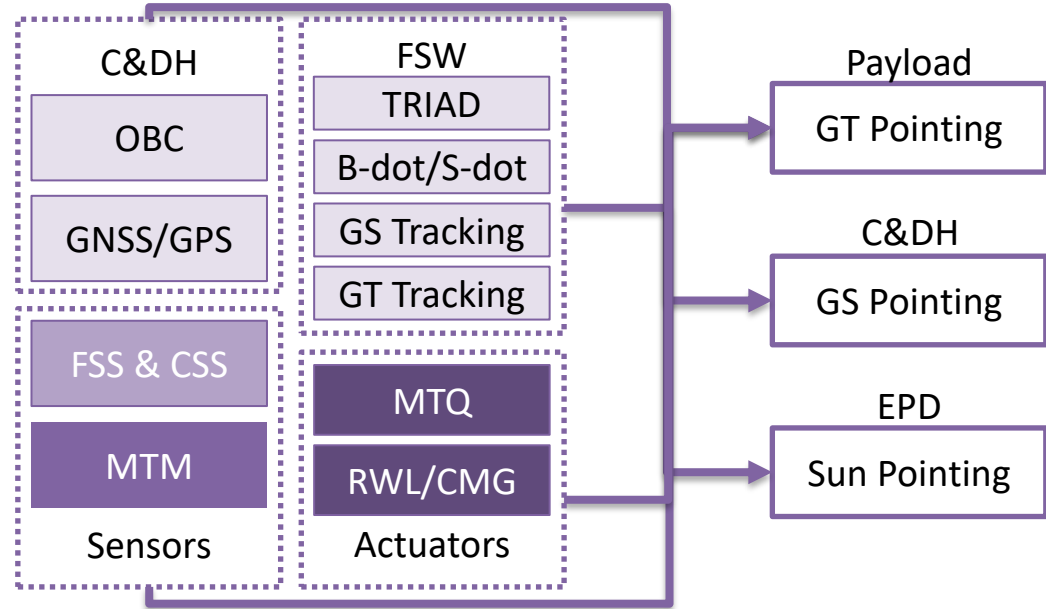
Phillip Shirkey, Logan Hayward, Michael Bennett,
Braden Adams, Sophia Fenton, Mitch Daues

Attitude Determination & Control

Driving Requirements

1. Ground Target Pointing (Sunlit imaging)
 - Includes nadir angular awareness
 - Highest precision requirement
2. Ground Station Pointing
3. Sun Pointing
4. Position and Time

Block Diagram

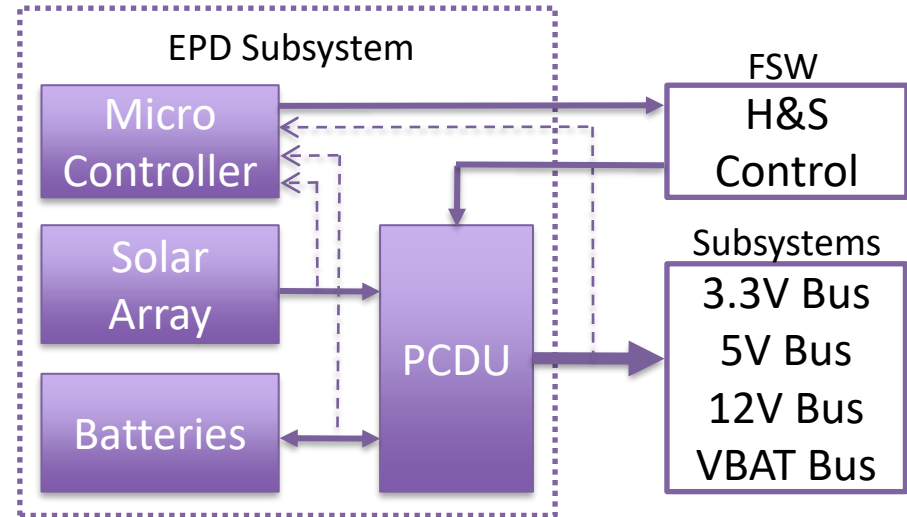


Electrical Power & Distribution

Driving Requirements

1. Remaining Power Positive
2. Ability to Test & Simulate
3. Ability to Monitor Current & Voltage

Block Diagram

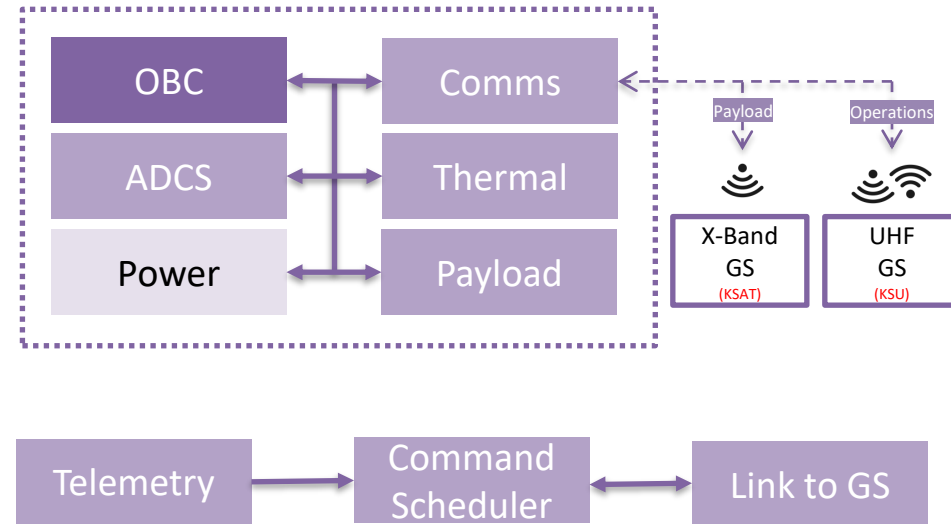


Communications & Data Handling

Driving Requirements

1. Payload Downlink
 - Number of images
 - Frequency of imaging
2. Operating Frequencies
 - X-Band (8.45-8.5 GHz)
 - UHF (435-438 MHz)
3. Communication Protocols
 - Interfacing, Networking, etc.

Block Diagram

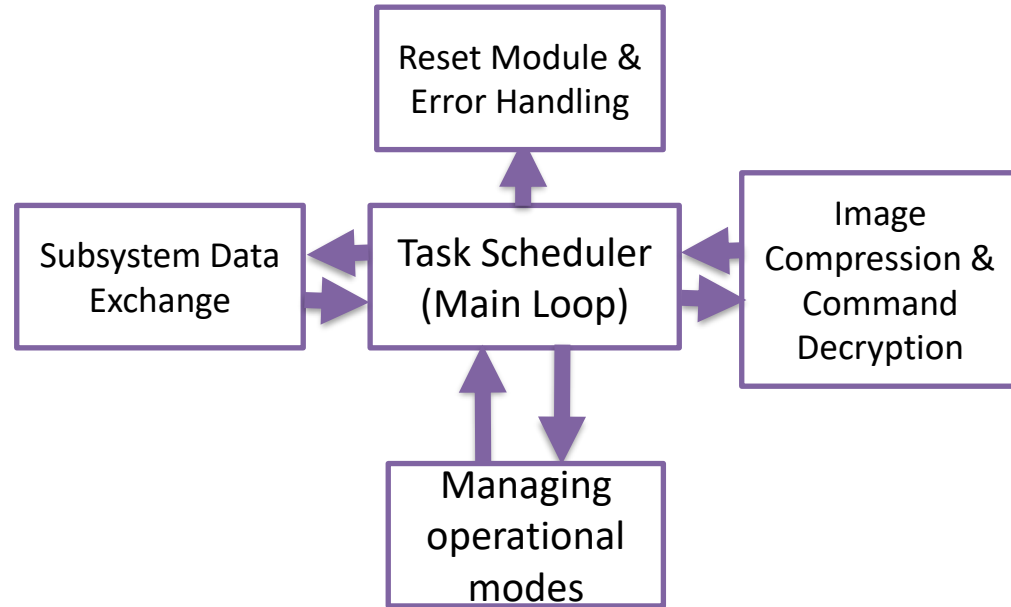


Flight Software

Driving Requirements

1. Respond to various operational modes
2. Collect and report subsystem telemetry data
3. Track date & time, location, and altitude for our images
4. Compress image data

Block Diagram

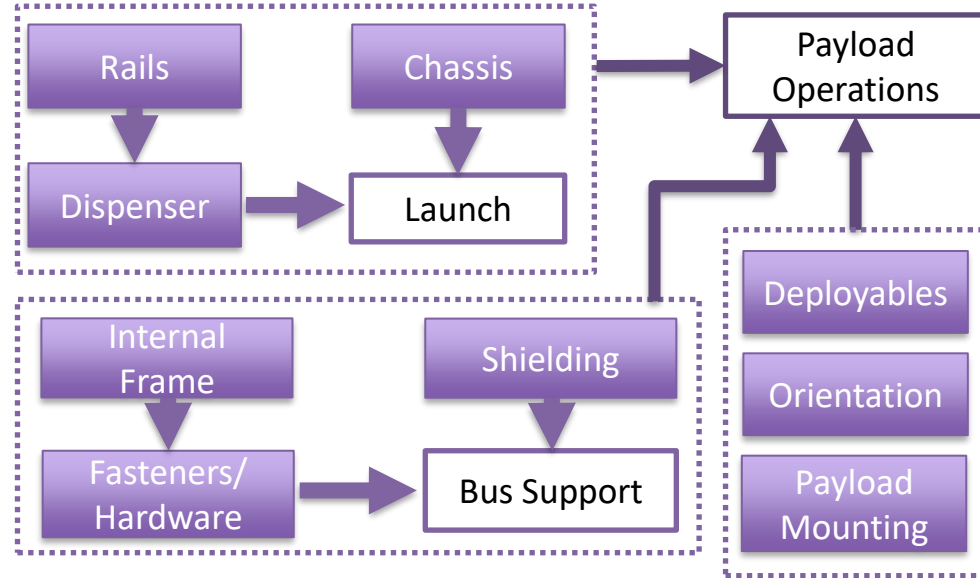


Structures

Driving Requirements

1. Launch Compatibility and Survivability
2. Satellite Bus Support and Component Integration
3. Payload and Mission Operations

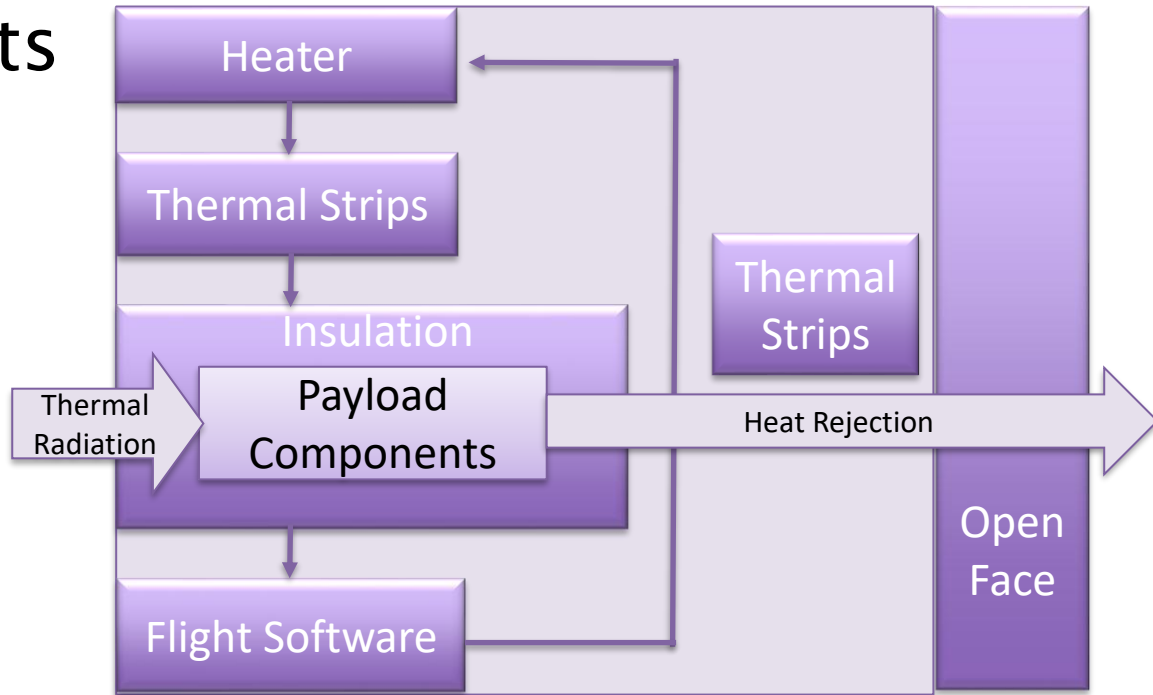
Block Diagram



Thermal

Driving Requirements

1. Maintain component operable temperature range
2. Maintain a stable/non-fluctuating temperature range (insulation, flight software, PID)





GROUND STATION

Noah Amundson, Andrew Osterhaus

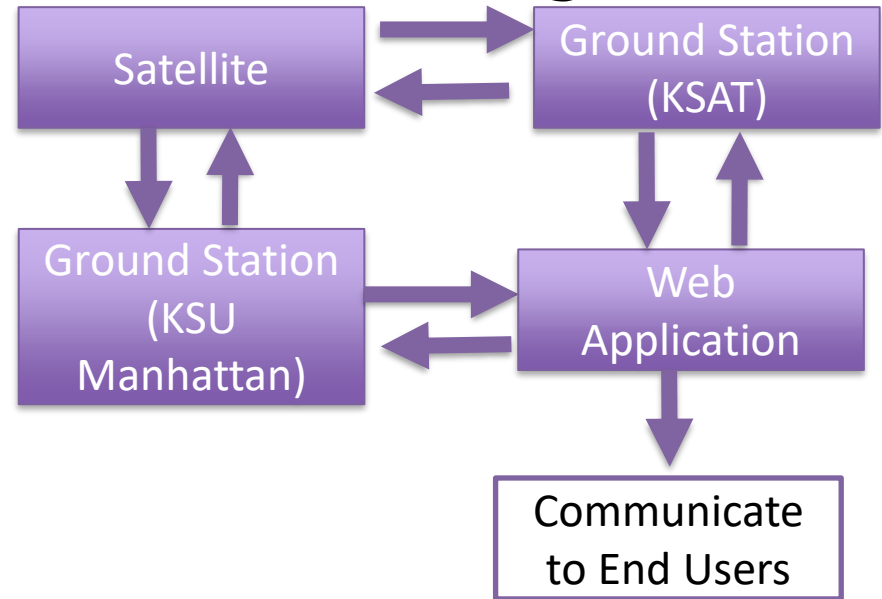
Ground Station Overview

Driving Requirements

1. Two ground stations required for mission

- UHF bands (430-440 MHz) for telemetry, tracking, and commands
- X-Band (8-12 GHz) for payload data

Block Diagram



Ground Station Hardware

Driving Requirements

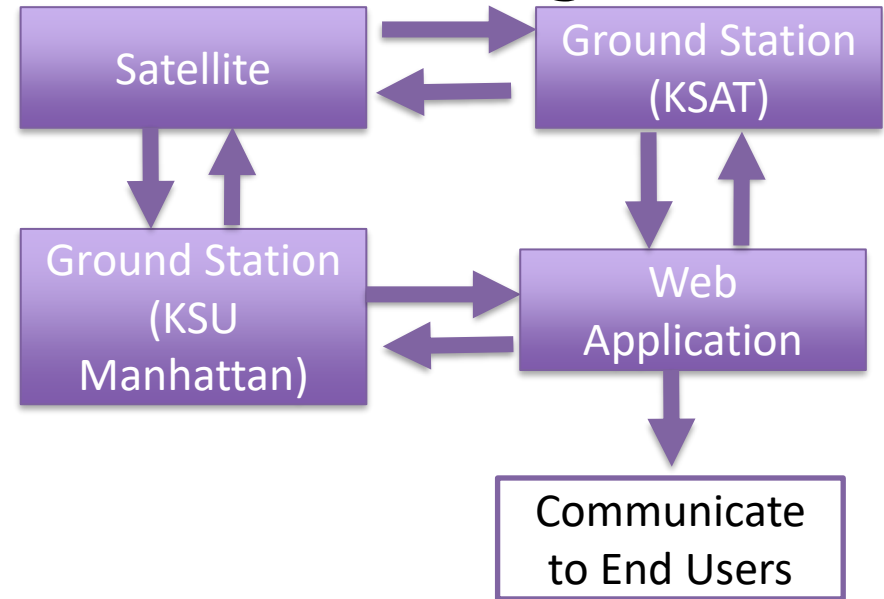
1. Location

- KSU Manhattan Radio Station
- KSAT (Kongsberg Satellite Services) in Norway

2. Licensing- FCC licenses for students.

3. Training- Students will need training to contact satellite.

Block Diagram

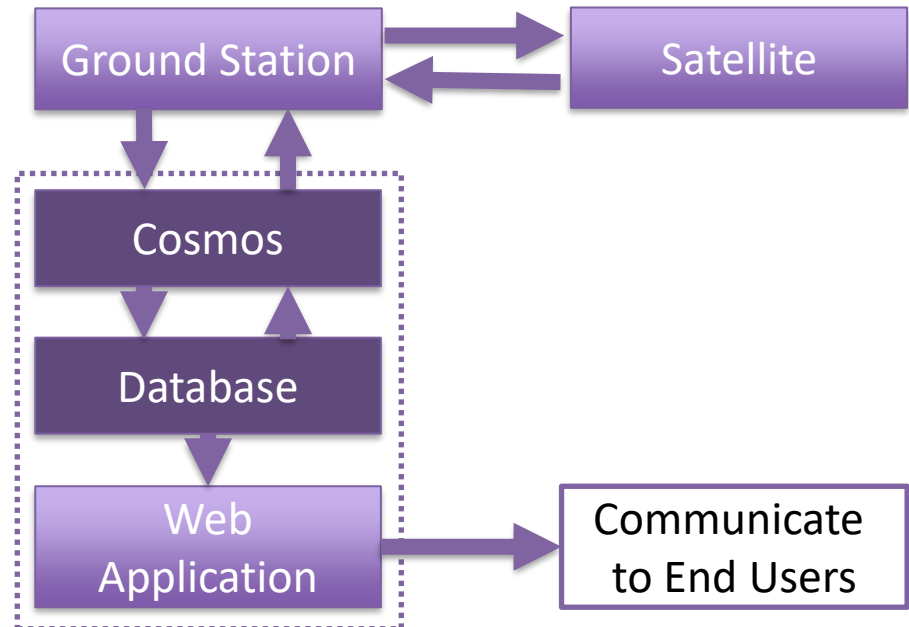


Ground Station Software

Driving Requirements

1. Communicating with the satellite
 - Receiving downlinked data
 - Up-linking data
2. Converting data to VI
3. Maintaining status of satellite
4. Distributing data to end-users

Block Diagram





MANAGERIAL

Chase Newell, Abigail Husted, Owen Franken

Integration and Testing

- RVM Verification Methods
 - Grouped by Analysis, Test, Inspect, Demonstrate (ATID)

Mission Parameter (Requirements)	Type	Description
System Compliance	D	Demonstrate that satellite and components meet set guidelines.
External Charging	T	Test the battery within system shows energy after being externally charged.
Radiation Hardness	A	Analyze components of ADC in a simulated software environment.
Optical Payload Mass	I	Inspect the payload mass with a scale to verify mass does not exceed "x" kg.

- 227 of 232 evaluated, expected completion Nov 14th, 2025
- MGSE/EGSE Equipment
 - Multimeter, power supply, function generator, oscilloscope, data acquisition system, RF-SCOE device, and grounding mat

Budget

- Payload = \$700,000 (\$500,000 = Simera Hyper-Scape 100)
- Comms = \$75,000
- Rest of Budget is similar to NuKat so far, overall goal currently is around \$1,000,000 for total hardware
- Launch = \$114,000

Comms Budget shown ->
 Orange = educated guess
 based on research
 Green = actual numbers from
 PHAT-Sat team members

Comms	Sub Category	Item	AH Guess L	AH Guess H	Item	Price
75,000	Radio Hardware	Transceiver	3,000	10,000	UHF Transceiver	7,900
	Radio Hardware	Power Amplifier	1,000	5,000	UHF Antenna	5,500
	Radio Hardware	Software Defined Radio	2,000	8,000	X-Band Transmitter	34,300
	Radio Hardware	Beacon Transmitter	500	1,500	X-Band Antenna	10,000
	Antenna System	Deployable Antennas	2,000	6,000	GNSS Receiver Antenna	15,000
	Antenna System	S-Band Patch Antenna	1,000	3,000		
	Antenna System	Antenna Deployment Mechanism	500	1,500		
	Antenna System	Antenna Coax & RF Harnessing	200	800		
	Supporting Electronics	RF Switch / Diplexer	200	1000		
	Supporting Electronics	Low-Noise Amplifier	500	1,500		
	Supporting Electronics	Grounding & EMI Filtering	100	400		
	Supporting Electronics	Power Conditioning Board	200	1,000		
	Onboard Data Handling	Interface				
	Onboard Data Handling	OBC Interface Board (if not integrated)	300	800		
	Onboard Data Handling	Software Integration / Telemetry Packets (low if work done in house)	500	200		
	Ground Segment Costs	Ground Station Access Fees	5,000	20,000		
	Licensing and Testing	FCC/ITU Frequency Licensing	1,000	5,000		
	Licensing and Testing	EMC/EMI Testing	3,000	5,000		
	Licensing and Testing	Thermal-Vac/Vibration Tests	5,000	15,000		
			LOW	HIGH	Real	
		GRAND TOTAL	26,000	85,700		72,700

Schedule

- Long Term: Mostly TBD, with more details to come with more research
 - Development – 12-24 months avg (here right now, 3 months in)
 - Launch contracting – 6-18 months avg
 - Integration – 3-9 months avg
 - Launch Queue – 3-12 months avg
 - **University CubeSat total– 18-24 months**

Short Term = Monday Project Management ->

PHAT-Sat ▾

Main table ... +

New item ▾ 🔍 Search 👤 Person ⚙️ Filter ▾ 📂 Sort ⚙️ Hide 📁 Group by ...

▾ Tier 1 (Mission)

☐	Item		Person	Status	Date
☐	CAD Model	✦ ⚙️	👤	Working on it	Dec 12
☐	Item 2	✦ ⚙️	👤	Done	Nov 11
☐	Item 3	✦ ⚙️	👤		Nov 11
☐	+ Add item				

▾ Tier 2 (LS, NukSat, ND, Bus, Gsys, GSE)

☐	Item		Person	Status	Date
☐	Research	✦ ⚙️	👤	Working on it	Nov 11
☐	Item 5	✦ ⚙️	👤		Nov 11
☐	+ Add item				

▾ Tier 3 (BO, CDH, FS, EPD, ADC, GStat, Therm, Struc)

☐	Item		Person	Status	Date
☐	Budget Details - Payload	✦ ⚙️	👤	Working on it	Nov 17
☐	CDH - PS RVM Review	✦ ⚙️	👤		
☐	Risk Mitigation Matrix	✦ ⚙️	👤	Working on it	Nov 12

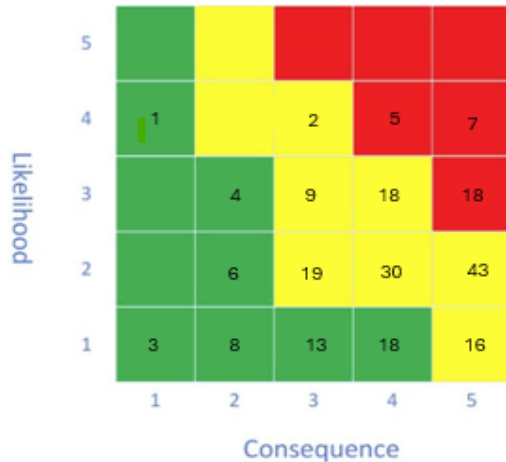
PHAT-Sat Competitive Analysis



Criteria	Weight	Phat-Sat		Drone(DJI Matrice 350RTK)		Other Satellite (PlanetScope)		People		Balloon Option		Aircraft	
		Score	Score with weight	Score	Score with weight	Score	Score with weight	Score	Score with weight	Score	Score with weight	Score	Score with weight
Cost	15%	3	0.45	4	0.6	1	0.15	5	0.75	4	0.6	3	0.45
Revisit Time	10%	2	0.2	2	0.2	5	0.5	1	0.1	1	0.1	2	0.2
Time to scan 1 arce (Speed)	30%	5	1.5	3	0.9	5	1.5	1	0.3	4	1.2	3	0.9
Capabilities	20%	5	1	5	1	4	0.8	4	0.8	3	0.6	3	0.6
Max Lifetime	10%	4	0.4	4	0.4	4	0.4	5	0.5	2	0.2	5	0.5
Key Issues	10%	3	0.3	4	0.4	4	0.4	3	0.3	2	0.2	2	0.2
Key Advantages	5%	5	0.25	5	0.25	4	0.2	3	0.15	1	0.05	5	0.25
	100%		4.1		3.75		3.95		2.9		2.95		3.1

Risk Assessment Strategy

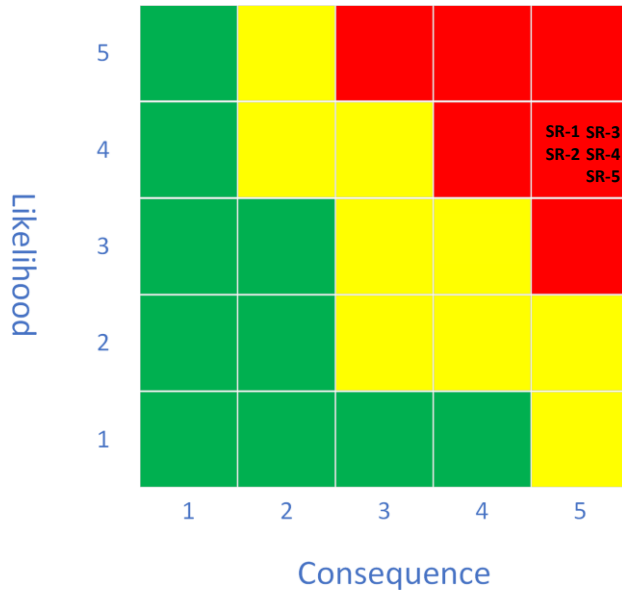
5x5 Matrix Number Breakdown of RVM



- Over 200 risks identified
- Focus on red level risks
- Working with sub teams to identify new potential risks
- Creating potential mitigation methods

24.09%
62.27%
13.64%

System-Level Risks



ID	Risk	Mitigation Strategy	Approach
SR-1	Satellite not receiving transferred information from ground software	Extensive testing to selected launch vehicle specifications and to run without transferred information for a period-of-time.	M
SR-2	EPD protection not sufficient from EMI and faults	Commence testing to simulate practical environment.	M
SR-3	The C&DH subsystem is unable to downlink "x" images to ground station.	Verify transfer capabilities and simulate impacts of disruptions, create secondary data transfer methods.	M
SR-4	FSW unable to control power distribution	Create system able to withstand LEO environment and verify reliability.	M
SR-5	FSW unable to receive and implement information	Design system to have backup data transfer capabilities.	M

CONCLUSION

Bailey Walke



Path to Design Review

Focus Area	PDR Goal
Science Needs	Develop science deliverables and stakeholder data needs. Research imager data processing and calibration processes
Orbit & ConOps	Finalize orbit selection, complete operations timeline, and verify feasibility with ground passes.
Requirements	Finalize system and subsystem requirements; ensure all are measurable and verifiable in the RVM.
Architecture & Interfaces	Freeze top-level architecture; define mechanical, electrical, and data interfaces.
Preliminary Subsystem Designs	Preliminary CAD, thermal models, power budgets, and link budgets.
Risk Reduction	Mitigate system-level risks, develop sub-system risks and assess
Verification & Test Planning	Outline environmental test plan per GEVS and identify required facilities.



MAB Feedback

- What mission definition content is missing?
- What mission definition needs more development?
- What mission definition trade studies are needed?

BACKUP

Electrical Power & Distribution (EPD)

Subsystem	Minimum Power Consumption	% of Operating Power for Small Spacecraft	% of Operating Power for Medium to Large Spacecraft
Payload	20 – 50 W	40	40 – 80
Propulsion	0 W	0	0 – 5
Attitude Control	0 W	15	5 -10
Communications	15 W	5	5 -10
Command and Data Handling	5 W	5	5 -10
Thermal	0 W	5	0 – 5
Power	10 – 30 W	30	5 – 25
Structure	0 W	0	0

- Example of Initial Power Budget
 - "New SMAD's Table 10-9", pg. 712, F. Zhu, *A Guide to CubeSat Mission and Bus Design*. Honolulu, HI: Pressbooks, 2022. Available: <https://pressbooks-dev.oer.hawaii.edu/epet302/>

Image Data Calculations

50% margins are used for a conservative estimate

$(18600 \text{ Mb image} / (6:1 \text{ compression} * 0.5) / (100 \text{ Mbps} * 0.5) / 60 \text{ seconds} \rightarrow \mathbf{2.1 \text{ min to downlink an image}}$

Simera HyperScape100 with 4.75 km swath $\rightarrow 837 \rightarrow 850$ images

$(850 \text{ images} * 2) * 2300 \text{ MB} \rightarrow \mathbf{3.9 \text{ Terabytes of storage}}$

Given 6-12 min of downlink time per pass we can estimate **3-5 images downlinked per pass**

To downlink the estimated 850 images, we would need between **170-284 passes**



Acronyms

- ADCS – Attitude Determination and Control System
- ATID – Analysis, Test, Inspect, Demonstrate
- B-dot/S-dot – Method of angular velocity determination using change in Earth or Sun vector orientation over time
- CAD – Computer Aided Design
- C&DH – Communications and Data Handling
- CMG – Control Moment Gyroscope
- ConOps – Concept of Operations
- CSS – Coarse Sun Sensor
- D.A.R.T – K-State Digital Agronomy Research Team
- EGSE – Electrical Ground Station Equipment
- EM – Electromagnetic
- EMI – Electromagnetic Interference
- EOSDIS – Earth Observing System Data and Information System
- EPD – Energy and Power Distribution
- FSW – Flight Software
- FWAHM – Full Width at Half Maximum
- FSS – Fine Sun Sensor
- GEVS – General Environmental Verification Standard
- GNSS – Global Navigation Satellite System
- GPS – Global Positioning System
- GS – Ground Station
- GSD – Ground Sampling Distance
- GT – Ground Target
- GSE – Ground Support Equipment
- HDPE – High Density Polyethylene
- H&S – Health and Status
- KSRE – K-State Research and Extension
- KSU – Kansas State University
- LEO – Low Earth Orbit
- LV – Launch Vehicle
- MAB – Mission Advisory Board
- MDR – Mission Definition Review
- MGSE – Mechanical Ground Station Equipment
- MS – Microsoft
- MSN – Main Subsystem Network
- MSND – Microstructured Semiconductor Neutron Detector
- MTM – Magnetometer
- MTQ – Magnetorquer (either air or solid core)
- NASA – National Aeronautics and Space Administration
- NDVI – Normalized Difference Vegetation Index
- NOAA – National Oceanic and Atmospheric Administration
- OBC – Onboard Computer
- PCDU – Power Control Distribution Unit
- PDR – Preliminary Design Review
- PID – Proportional-Integral-Derivative (controller)
- RDT – Radiation Detection Technologies
- RF-SCOE – Radio Frequency Special Check-Out Equipment
- RGB – Red Green Blue
- RTC – Real Time Clock
- RTD – Resistance Temperature Detector
- RVM – Requirement Verification Matrix
- RWL – Reaction Wheel
- Rx – Receive
- SAA – South Atlantic Anomaly
- SDP – Standard Data Product
- SI – Soil Index
- SR – System Risk
- TBD – To Be Determined
- TRIAD – Tri-Axial Attitude Determination
- Tx – Transmit
- UHF – Ultra High Frequency
- VI – Vegetation Index